Lakshya Tiwari

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Summary

Backend and AI Engineer with experience building Node.js services, distributed workflows, and modular applications in Turborepo monorepos. Contributes to LLM-powered applications using MCP servers, RAG pipelines, Redis-backed worker queues, and Docker.

Education

JSS Academy of Technical Education Noida, India
B.Tech. in Information Technology 2023 – 2027 (Expected)
CGPA: 8.54/10

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C++, SQL
Frontend & Authentication: React.js, Next.js, Tailwind CSS, JWT, OAuth, Clerk
Backend: Node.js, Express.js, FastAPI, REST APIs, Redis, Celery, Worker Queues, Async Job Processing
AI Engineering: LangChain, LangGraph, RAG (Naive, Adaptive, Corrective, Hybrid), MCP Servers, Mem0
Databases & Retrieval: PostgreSQL, MongoDB, Prisma, Mongoose, Qdrant (Vector DB), Neo4j (Graph DB)
Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions (CI/CD), Git, Linux, Vercel
Core Subjects: Data Structures, Algorithms, Computer Networks

Experience

Cognify Systems April 2026 – November 2026
Full Stack AI Engineer Intern Remote

- Engineered Node.js backend services and contributed to distributed workflows on Redis-backed worker queues, decoupling long-running jobs from API request handling.
- Contributed to a modular Turborepo monorepo, using shared packages for code reuse across applications.
- Collaborated with the engineering team to deliver MCP servers and 4 RAG architectures: Naive, Adaptive, Corrective (CRAG), and Hybrid, enabling distinct retrieval workflows.
- Developed 2 core pipelines ie ingestion and retrieval pipelines linking backend services to LLM-powered workflows, enabling document processing and context retrieval for AI applications.
- Containerized services with Docker (Dockerfiles and configurations) to standardize development and deployment environments for the team.

Projects

Cortex: GitHub Repository Q&A Assistant GitHub
Tech Stack: Next.js, Python, Celery, LangChain, LangGraph, LangSmith, Mem0, Qdrant, Neo4j, Docker

- Built a web platform for repository Q&A, using RAG to answer questions about code architecture, data flow, and source files by connecting github repository.
- Implemented AST-based hierarchical code chunking using Tree-sitter to preserve function and class boundaries, enabling structure-aware embedding generation and more contextually relevant code retrieval.
- Designed a 3-component decoupled architecture: application services enqueue ingestion jobs, Celery workers process repositories, and a separate Python API serves retrieval requests.
- Combined 2 retrieval backends, Qdrant vector search and Neo4j graph retrieval, with Mem0 conversational memory, LangSmith tracing, and Dockerized local services.

KeyCrypt: Secure Password Vault GitHub | Live
Tech Stack: Next.js, TypeScript, Web Crypto API, Supabase, PostgreSQL, Prisma

- Shipped a live, browser-based password vault supporting 50+ credentials with client-side AES-GCM encryption and PBKDF2 master-key derivation.
- Enforced zero plaintext credential storage on the server by running encryption and decryption client-side with the Web Crypto API.
- Integrated Supabase Authentication, email verification, and PostgreSQL persistence through Prisma.

Achievements

- **LeetCode:** Peak contest rating of 1780+; solved 500+ problems across data structures and algorithms.
- **Hackathon Wins:** Won multiple college-level hackathons in teams, delivering prototypes in 24–36 hours.
- **Esports:** Won college-level PUBG tournaments.